



# Aakash Avadhut Dhekane

CV

## About me

AI & ML Engineer / Senior Data Scientist with 8+ years of experience turning research driven ideas into production grade data and AI systems. I work end-to-end across the lifecycle – from ideation, prototyping, and experimentation to deployment, monitoring, and continuous improvement with a strong focus on measurable business impact.

Currently at Shopware AG, I lead the development of GenAI and agentic AI solutions, integrating large language models with robust backends and MLOps practices. In previous roles at HeyJobs, Talpasolutions, and trivago, I delivered classical machine learning solutions in marketing, sensor analytics, and recommender systems.

I enjoy acting as a bridge between business and technology: facilitating discussions and acting as a trusted consultant for stakeholders (including C-levels), clarifying problem statements, and turning them into actionable data and AI use cases.

## Skills

Python, PyData Stack, SQL  
XGBoost, LightGBM, CatBoost,  
PyTorch  
FastAPI, Redis, PySpark  
Docker, Kubeflow, Sagemaker,  
Terraform  
Github and Github Actions  
AWS & related services  
Langchain, Langgraph  
RAG, Agentic AI  
Statistical analysis, predictive  
modeling  
Machine Learning, A/B Testing

## Personal details

- 🏠 Bankstr. 73  
40476 Düsseldorf
- 📍 Germany
- 📞 017657700763
- ✉ aakash.unlimited@gmail.com

## Professional experience

**Shopware AG**  
Senior Data Scientist

**06/2024 - Present**

- Spearheaded the end-to-end development of a GenAI powered Merchant Assistant, delivering conversational insights for thousands of e-commerce users and embedding it into core workflows.
- Led integration of Claude (via AWS Bedrock) and Mistral (Pilot 2502) with LangChain/LangGraph, enabling tool based reasoning, agentic orchestration, and scalable RAG pipelines.
- Architected a production grade backend using FastAPI, Docker, and Terraform on AWS (EKS, Lambda, CloudWatch), ensuring CI/CD readiness, monitoring, and compliance for enterprise use.
- Drove crossfunctional GenAI adoption by aligning product, engineering, and business stakeholders, and helping them prioritize use cases based on feasibility and impact.
- Founded the 'AI Papers Discussion Group', a weekly internal forum to review and debate cutting edge AI research papers relevant to Shopware's roadmap. This initiative led to the implementation of structured logging for training Small Language Models
- Mentored two data scientists in MLOps and AI ethics for scalable enterprise deployment.

## Educational background

---

### University of Koblenz-Landau

2014 - 2017

Masters of Science

Web Science (Renamed as Data Science)

### Heyjobs GmbH

05/2023 - 05/2024

Senior Data Scientist

- Reduced marketing spend by  $\approx 10\%$  via ML models predicting application quality and early stopping criteria across channels.
- Co-defined product vision and re-designed KPIs for revenue maximization, creating an execution plan and architecture to enhance Marketing Mix Models for budget optimization.
- Acted as the bridge between business, data, and engineering, ensuring consistent alignment of technical outcomes with corporate KPIs.
- Initiated the shift from fixed to data-driven attribution modeling, aligning Marketing, Finance, and Data teams around shared metrics.

### Talpasolutions GmbH

12/2020 - 04/2023

Data Scientist

- Developed, maintained, and enhanced FTI pipelines for ML models (Kubeflow on AWS).
- Analyzed time-series sensor data (Python, pandas, XGBoost) for activity detection and KPI generation for 500+ machines.
- Led company-wide validation and data labeling strategy, implementing a validation pipeline (AWS SageMaker GroundTruth).
- Collaborated with product and domain experts to define meaningful KPIs and ensure models addressed real operational needs.

### trivago n.V.

04/2017 - 11/2020

Data Scientist

- Improved CTR by 2% by modeling user intent (pricing indexes) and personalizing listings/rankings, impacting customer value.
- Increased user interaction/content touchpoints by 5% via a filter recommender engine (Python, Hive, LightGBM).
- Reduced regression testing time by 50% using clustering for representative datasets and developing a regression testing module (F1-score, MRR).
- Developed ML training/inference pipelines and monitored experimentation (A/B Testing & Looker dashboards).